

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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Abstract

001 Aggregate spatial accuracy can hide sharply differ-
002 ent capabilities within vision–language models (VLMs).
003 We study object-relative orientation as a diagnostic
004 case. On Visual Spatial Reasoning (VSR), moving from
005 a compact 2.2B model to a larger 7B VLM substan-
006 tially improves overall, depth, horizontal, and contain-
007 ment accuracy, but orientation changes only modestly.
008 Except for structured prompting, which degrades per-
009 formance, trained conditions on both models remain
010 in a narrow 62–66% orientation band. We test struc-
011 tured prompting, general and targeted LoRA, hard neg-
012 atives, visual/projector adaptation, frozen and object-
013 grounded representation probes, and object-centric de-
014 composition; none of the tested configurations reliably
015 closes the gap, and a single-annotator clean-label sen-
016 sitivity analysis shows that annotation ambiguity does
017 not explain it away.

018 Accuracy also fails to capture relational coherence.
019 LM-side adaptation significantly improves consistency
020 over zero-shot, while hard-negative training raises
021 facing/facing-away consistency from 66.0% to 77.7%
022 with facing accuracy unchanged at 68.9%; the addi-
023 tional pooled consistency gain over LM-only is not sig-
024 nificant. Cross-dataset evaluation on SITE shows that
025 VSR-trained adaptation does not transfer cleanly: it is
026 neutral overall and significantly harms official spatial-
027 relationship reasoning (75.1%→71.6%, $p = 0.004$). A
028 preregistered orientation-vocabulary slice is analyzed
029 as exploratory evidence rather than as direct replica-
030 tion of the VSR construct. The results point to a
031 persistent orientation bottleneck in which orientation
032 is not robustly decodable by the tested frozen-feature
033 readouts and relational decisions are not always propa-
034 gated coherently, while also showing that task-specific
035 spatial adaptation can transfer poorly.

1. Introduction

Spatial reasoning is central to multimodal systems that must interact with physical environments. A useful VLM should not only recognize objects, but determine whether one object is in front of another, whether it is facing another, whether two objects are parallel, and whether equivalent or complementary formulations yield logically compatible answers. These abilities matter for robotics, embodied agents, navigation, and scene understanding.

Existing work already shows that spatial relations are difficult for vision–language systems. VSR was designed as a controlled natural-image probe and reported particularly poor performance on object-orientation relations in earlier VLMs [7]. More recent controlled benchmarks likewise find that strong VLMs can fail on basic left/right and other grounded relations [6, 9], while broader evaluations continue to identify spatial orientation as a difficult factor [14]. Our question is therefore not whether spatial reasoning can fail, but whether the orientation weakness persists in modern generative VLMs of different sizes and after targeted intervention, and what the failure signature reveals about representation, reasoning, and transfer.

In our VSR experiments, the distinction between relation families is striking. Moving from SmolVLM2-2.2B-Instruct to the larger Qwen2-VL-7B-Instruct increases zero-shot overall accuracy from 74.0% to 80.9%. Horizontal reasoning rises from 70.2% to 84.8%, depth from 68.9% to 75.2%, and containment from 83.4% to 89.3%. Orientation, by contrast, moves only from 62.8% to 63.5%. We then evaluate interventions spanning prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and explicit object-centric decomposition. Structured prompting significantly hurts performance; general and targeted LoRA improve aggregate accuracy but do not remove the orientation ceiling; hard negatives improve targeted relational coherence without a

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075 significant overall orientation-accuracy gain; and the
076 tested visual-side LoRA configurations do not outper-
077 form the language-side control.

078 As a sensitivity analysis, removing examples flagged
079 by a single-annotator audit raises orientation accuracy
080 (65.7%→78.5% for 7B General LoRA, 66.4%→76.6%
081 for hard-negative LoRA on the strictest 107-example
082 subset) but does not eliminate error (2B structured
083 shows no gain). Because the removed examples were
084 drawn from persistent failures, part of this gain is ex-
085 pected by construction.

086 A second failure mode emerges from logical consis-
087 tency. The zero-shot 7B model is consistent on only
088 36.9% of complementary facing/facing-away pairs. LM-
089 side LoRA substantially improves coherence, and hard-
090 negative training raises facing consistency to 77.7%
091 while facing accuracy remains unchanged at 68.9%.
092 The extra hard-negative gain is not significant in the
093 pooled strict-family comparison, yet accuracy stays un-
094 changed while consistency moves: the two metrics cap-
095 ture different behavior.

096 Finally, we test cross-dataset transfer. The 7B
097 model is strong on SITE’s official spatial-relationship-
098 reasoning category (75.1% raw; 59.2% chance-adjusted
099 accuracy), yet the VSR-trained adapter does not
100 generalize cleanly: it is statistically neutral overall
101 (54.2%→53.8%, $p = 0.66$) and significantly degrades
102 the official category (75.1%→71.6%, $p = 0.004$). A
103 preregistered, non-official orientation-vocabulary slice
104 is much weaker in the aggregate (48.3% raw; 24.0%
105 chance-adjusted) and shows no positive transfer, but—
106 as we detail below—its composition limits what it can
107 establish, and we therefore do not treat SITE as inde-
108 pendent confirmation of the VSR orientation construct.

109 Our contributions are fourfold:

- 110 • Fine-grained diagnosis across two differently sized
111 VLMs. Across ten VSR conditions and two dif-
112 ferently sized models, several spatial families im-
113 prove substantially while object-relative orientation
114 remains unusually resistant.
- 115 • Robustness and mechanism analysis. A single-
116 annotator clean-label sensitivity analysis,
117 frozen/object-grounded probes, visual-side adapta-
118 tion, and object-centric decomposition constrain
119 purely annotation-based or single-component
120 explanations.
- 121 • VSR-specific accuracy-coherence separation.
122 Complementary-relation tests on VSR’s strict
123 relation pairs show that, for the tested conditions,
124 relational consistency can change substantially
125 while task accuracy remains fixed.
- 126 • Cross-dataset transfer analysis. SITE shows strong
127 official spatial-relationship performance, while VSR-

trained LoRA transfers poorly and significantly
harms the official SITE category; the orientation-
vocabulary slice is analyzed as exploratory evidence.

2. Related Work 131

Spatial reasoning benchmarks. VSR introduced a con-
trolled natural-image benchmark of more than 10k
image-text pairs spanning 66 spatial relations and ex-
plicitly highlighted orientation relations such as fac-
ing and parallel as difficult for the VLMs studied at
the time [7]. Our work should therefore be read as a
modern diagnostic extension rather than the first ob-
servation of an orientation weakness. What’sUp iso-
lates basic spatial relations by holding object identi-
ties fixed while varying configurations, and found large
gaps between humans and tested vision-language mod-
els [6]. GSR-Bench extends controlled grounded spa-
tial evaluation across a larger set of multimodal mod-
els and model scales [9]. SITE broadens evaluation
to single-image, multi-image, and video inputs and or-
ganizes tasks using cognitive-science dimensions includ-
ing spatial visualization/orientation, intrinsic/extrinsic
relations, and static/dynamic settings; its authors re-
port spatial orientation as a persistent challenge [14].
Recent broad evaluations such as SpatialScore simi-
larly argue that spatial intelligence remains fragmented
across tasks and modalities [15], and Mind the Gap
isolates spatial reasoning from detection and semantic
comprehension in contemporary VLMs [10].

Diagnosing spatial failures. Recent work has begun
to analyze spatial failures mechanistically rather than
only benchmarking them. On VSR itself, AdaptVis
traces attention distributions and finds that success-
ful spatial reasoning correlates with attention align-
ing to actual object locations [2]. CREG charac-
terizes the directional organization of attribution ev-
idence in Qwen2-VL on spatial-relation benchmarks,
showing that standard heatmaps often reflect image
layout rather than the queried relation [11]. When
More Is Less analyzes when injecting spatial cues, com-
monsense knowledge, or chain-of-thought instructions
helps or hurts VSR, with effects varying by model and
setup [1]. Probing Visual Concepts measures linear
decodability of visual concepts, including orientation-
relevant ones, in lightweight driving VLMs and identi-
fies information-flow bottlenecks [12]. VSP decomposes
visual spatial planning into perception and reasoning
subtasks [16], and MM-Spatial targets 3D spatial un-
derstanding with geometric inputs [3].

Prior work has thus already established spatial and
orientation weaknesses in VLMs and has begun to ana-
lyze those failures through prompting, attention, repre-

179	sentation probing, and consistency-based training. Our	227
180	contribution is therefore not the first identification of	228
181	orientation difficulty or spatial inconsistency. Instead,	
182	we revisit the known VSR orientation weakness in mod-	
183	ern generative VLMs and subject the same relation fam-	
184	ily to a controlled diagnostic ladder spanning adapta-	
185	tion, frozen and object-grounded readouts, visual-side	
186	interventions, complementary-relation consistency, label	
187	sensitivity, and cross-dataset transfer.	
188	Adaptation and spatial specialization. We use LoRA,	
189	which adapts frozen models through trainable low-	
190	rank updates [5], to compare general, targeted, hard-	
191	negative, projector, and vision-plus-projector interven-	
192	tions. This design lets us ask whether benchmark	
193	gains reflect general spatial competence or specializa-	
194	tion. The external SITE transfer experiment is central	
195	here: VSR adaptation is neutral overall on SITE,	
196	does not improve the orientation subset, and signifi-	
197	cantly degrades SITE's official spatial-relationship cat-	
198	egory. The result complements benchmark work by	
199	explicitly measuring negative as well as positive cross-	
200	dataset transfer.	
201	Representation probing and logical consistency. Di-	
202	agnostic probes can reveal whether information is ac-	
203	cessible from frozen representations, but probe perfor-	
204	mance must be interpreted cautiously because probe	
205	capacity can itself learn the target task [4]. We there-	
206	fore use linear and nonlinear probes, compare against	
207	majority baselines, add object-grounded features, and	
208	avoid claiming that weak decodability proves informa-	
209	tion is absent. Separately, consistency across trans-	
210	formed or complementary statements has prior litera-	
211	ture, including consistency-oriented training objectives	
212	such as SAGE's geometric and linguistic duality oper-	
213	ations [8]. Our consistency analysis is a VSR-specific	
214	diagnostic rather than a new consistency framework:	
215	it uses strict complementary relation pairs with ob-	
216	served verdict-pattern and base-rate analysis, and it	
217	shows that instance-level accuracy and complementary-	
218	relation coherence can diverge for the tested conditions.	
219	3. Experimental Setup	
220	3.1. Benchmarks	
221	VSR. We use VSR [7] as the primary diagnostic	
222	benchmark. The random-split test set contains 2,195	
223	examples. The benchmark defines 64 relation labels;	
224	61 appear in the evaluated test split. The orientation	
225	family contains facing, facing away from, parallel to,	
226	and perpendicular to, totaling 137 test examples. All	
	main VSR results use the same held-out test set and	227
	deterministic True/False parsing.	228
	SITE. We use SITE [14] for cross-dataset transfer	229
	evaluation only and do not train on SITE. The image	230
	evaluation contains 2,591 examples: 1,368 single-image	231
	and 1,223 multi-image examples. Our primary SITE	232
	subset is the official spatial-relationship-reasoning cate-	233
	gory ($n = 993$). We additionally report a transparently	234
	defined, non-official keyword-derived orientation subset	235
	($n = 2,024$), analyzed as exploratory evidence rather	236
	than as direct replication of the VSR construct. SITE	237
	results use the benchmark's native multiple-choice for-	238
	mat and chance-adjusted accuracy (CAA) in addition	239
	to raw accuracy.	240
	3.2. Models and Conditions	241
	Our compact model is SmolVLM2-2.2B-Instruct, and	242
	our larger model is Qwen2-VL-7B-Instruct [13]. We	243
	evaluate ten canonical VSR conditions: 2B zero-shot,	244
	2B structured prompting, 2B General LoRA, 2B Tar-	245
	getted LoRA, 7B zero-shot, 7B General LoRA, 7B Tar-	246
	getted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA,	247
	and 7B Vision+Projector LoRA. LoRA [5] is used for	248
	all parameter-efficient adaptation conditions.	249
	General LoRA uses a proportional sample across re-	250
	lation families. Targeted LoRA allocates most training	251
	examples to weak orientation/depth/horizontal fam-	252
	ilies with rehearsal from the remaining relations. Hard-	253
	negative LoRA introduces audited complementary ori-	254
	entation examples, primarily facing/facing-away inver-	255
	sions. Vision-side conditions adapt either the multi-	256
	modal projector or upper vision blocks together with	257
	the projector. The frozen experimental tag paper-	258
	freeze-v1 records the final result set.	259
	3.3. Metrics and Statistical Tests	260
	For VSR we report accuracy overall and by spatial fam-	261
	ily. For SITE we report raw accuracy and CAA. Paired	262
	condition comparisons use McNemar tests when ap-	263
	plicable. Clean-label orientation analysis reports the	264
	full orientation set and progressively stricter audited	265
	subsets. Logical consistency is evaluated by construct-	266
	ing complementary statements for strict relation pairs	267
	(left/right, front/behind, facing/facing-away) and test-	268
	ing whether the model assigns opposite verdicts as re-	269
	quired. Parallel/perpendicular is treated separately as	270
	a soft complement because both relations can be false	271
	for oblique objects.	272

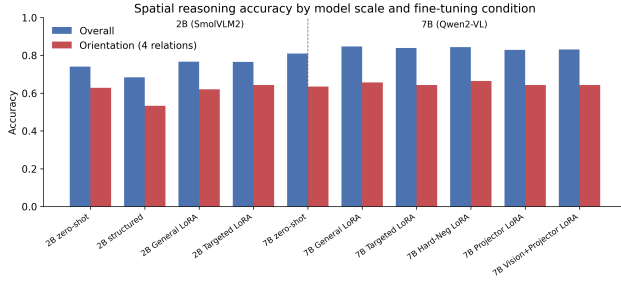


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

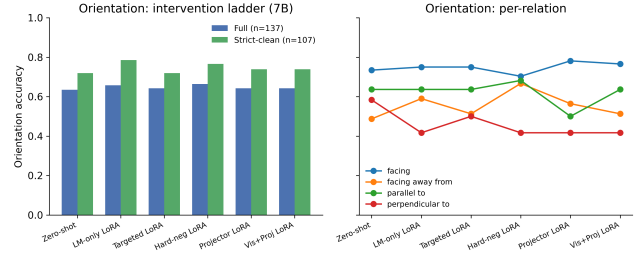


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band.

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent analyses of information injection on VSR, where additional spatial cues or reasoning instructions can help or hurt depending on the model and setup [1], this result is specific to the tested 2B condition and prompt; it does not establish that prompting generally fails.

4.2. Clean-Label Sensitivity Analysis

A single-annotator exploratory audit identified ambiguous or questionable orientation examples, so we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these values should not be read as a matched comparison against other relation families. The robust conclusion is narrower: removing flagged examples raises orientation scores but does not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations without fine-tuning the generative model. Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the confidence interval overlaps the majority baseline. In short: orientation is not robustly decodable by the tested frozen-feature readouts.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Linear decodability of visual concepts has been probed in driving-oriented VLMs [12], and directional information in this model family has been inspected via attribution [11]; our probe analysis is not the first of either kind. It differs by targeting VSR’s orientation relation family with pooled, patch-vote, and object-grounded readouts over the exact conditioning features of the generative model, and by pairing decodability with adaptation and consistency interventions rather than treating it as an end in itself. Detailed probe tables and plots are in the supplementary.

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not unlock the ceiling.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s True/False verdicts agree with the end-to-end control on only 55.1–58.3% of the 127 boxed orientation statements; in the committed analysis (10 no-box statements scored wrong), verdicts diverge on 61–65 statements, and the control is correct while the pipeline is not in 44–47 of them. The committed pipeline scored agreement with the control rather than ground-truth accuracy, so we report the tested decomposition as failing to reproduce the generative model’s behavior, not as a direct accuracy comparison. Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair and transformation-based consistency evaluation has prior literature, including consistency-oriented training [8]; our contribution is the VSR-specific diagnostic below, built on strict complementary relation pairs with observed verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The observed verdict patterns show what drives this: the model affirms either member of a facing pair only 17–20% of the time, a False-defaulting bias that yields both-False verdicts on 63.1% of pairs and no both-True verdicts (0/103); expected consistency under verdict independence given these base rates is 30.2%, below the observed 36.9% (verdict table in the supplementary): the dominant phenomenon is a refusal to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind consistency to 70.7%; pooled strict-family consistency is significantly better than zero-shot ($p < 0.0001$). Hard-negative training further raises facing consistency to 77.7%. Facing accuracy, however, is identical for LM-only and hard-negative LoRA at 68.9%. The additional hard-negative improvement is not significant in the pooled strict-family McNemar comparison ($p = 0.29$). Paired hard-negative supervision therefore provides a targeted intervention on complementary-relation coherence: it descriptively raises facing-pair consistency while leaving original facing accuracy unchanged, although the additional

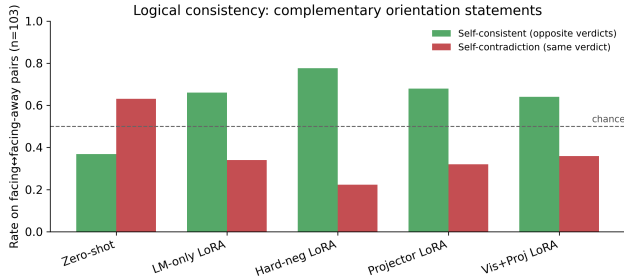


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

pooled consistency effect over LM-only is not significant. We do not claim a general coherence algorithm; the key result is separability: identical facing accuracy coexists with substantially different facing consistency. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [14], with no SITE training. Because the preregistered decision rule resolved to branch 2 (the official category was not broadly weak), this adapter run is a post-protocol cross-dataset transfer check rather than a preregistered step; the protocol and its outcome are documented in the supplementary. The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer on the independent benchmark; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The preregistered, non-official orientation-vocabulary slice is substantially weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category, and below the 54.2% / 31.1% all-images average) and shows no positive transfer under the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

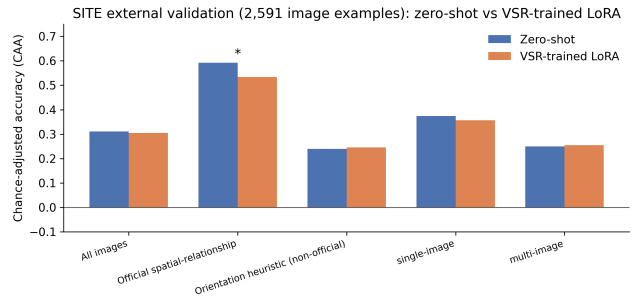


Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

category), we treat it as exploratory evidence rather than as direct replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI [0.60, 1.16], $p = 0.29$), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation that orientation relations are

486 difficult [7], while extending it to modern generative
487 VLMs and a much broader intervention analysis.

488 Decodability and coherence both matter, as diagnos-
489 tics. The probes show that orientation is not robustly
490 decodable under the tested frozen-feature readouts,
491 while consistency experiments expose large logical fail-
492 ures even for relations answered above chance. Neither
493 result alone identifies a complete causal mechanism,
494 and both are diagnostic rather than causal-localization
495 claims. Together they support a mixed account in
496 which orientation information is difficult to access reli-
497 ably under the tested readouts and relational decisions
498 are not always propagated coherently across comple-
499 mentary formulations.

500 Task adaptation can specialize rather than general-
501 ize. General LoRA improves VSR overall accuracy
502 substantially, but its SITE transfer is neutral over-
503 all and significantly negative on the official spatial-
504 reasoning subset. Hard negatives improve a targeted
505 coherence behavior while producing relation-specific
506 tradeoffs. Benchmark-specific fine-tuning gains should
507 therefore not be equated automatically with general
508 spatial-reasoning improvement.

509 8. Limitations

510 Our main diagnostic test contains only 137 orientation
511 examples, and individual relation subsets such as per-
512 pendicular are smaller still ($n=12$); perpendicular re-
513 sults are therefore reported descriptively throughout.
514 The single-annotator clean-label audit reduces ambigu-
515 ity but also decreases sample size, and it is exploratory
516 qualitative evidence rather than a ruling on label noise.
517 The SITE orientation slice is keyword-derived rather
518 than an official benchmark category; a composition-
519 confound analysis shows its aggregate score is largely
520 explained by task composition, so we do not treat SITE
521 as independent confirmation of the VSR orientation
522 bottleneck. Our probing methods test accessible decod-
523 ability under specific linear/MLP and pooling choices;
524 failure to decode does not prove information is absent.
525 Visual-side experiments use LoRA rather than full end-
526 to-end retraining. We study two differently sized VLMs
527 from different families; this is a cross-model compar-
528 ison, not a controlled scale experiment. All trained
529 conditions are single-checkpoint runs: the paired Mc-
530 Nemar tests quantify disagreement on the fixed test
531 examples, not training-run variability, and these single-
532 seed results do not estimate it. The SITE evaluation is
533 image-only; SITE video reasoning remains future work.
534 Finally, several interventions are evaluated as negative

results, so lack of improvement should be interpreted
for the tested configurations rather than as impossibil-
ity claims about all prompting, adaptation, or architec-
tural methods.

9. Conclusion

Aggregate spatial accuracy hides a persistent orien-
tation bottleneck in the VLMs studied here. A
larger VLM and standard adaptation substantially
improve several spatial families while leaving object-
relative orientation comparatively resistant. Auditing,
probing, visual-side adaptation, and decomposition
constrain simple explanations, while complementary-
relation tests show that accuracy and coherence are
distinct axes of spatial competence. Cross-dataset eval-
uation on SITE further shows that VSR-specific adap-
tation transfers poorly and can negatively transfer to
official spatial-relationship reasoning. The results mo-
tivate spatial evaluation that reports relation-specific
performance, transfer, and logical coherence rather
than relying on aggregate accuracy alone.

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